Factor-base shape leads

Native exploratory search · 2026-10-18

The requested search pursued structures that preserve useful successful decompositions and independent relations. It did not search for cheap rejection alone. Native sweeper source b3ed8889 ran the frozen public-fixture plans; a separate Rust field/group implementation then exhaustively audited their two-point support. No scalar logarithms were recovered.

Main observed lead

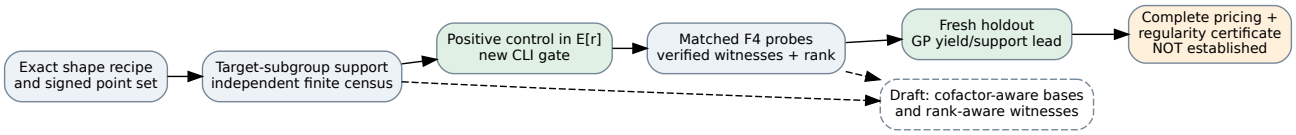
On **icv1-f2m13-t181-515ee569**, trace-zero geometric progression (GP) bases improve supported-target coverage in this finite panel. Degree13 is secondary evidence; the degree 31 primary results are described below.

Constructor	Cols	Holdout /128	Coverage /2002	Train rank
Coordinate	16	4	144	5
Random seed 17	12	4	78	3
Random seed 29	17	13	152	2
Trace-zero GP seed 17	15	15	226	5
Trace-zero GP seed 29	15	17	214	12
Trace-zero GP seed 43	21	30	482	15

The same 128 fresh draws contain 122 distinct targets. GP seeds 17/29 have 15 distinct successful holdout targets each; GP seed 43 has 30. Different inventory sizes are explicit. These are observed counts and exact finite support, not a significance test, a total-cost ranking, or a fastest result. Successful observed solving degree remains 3; no regularity improvement is established.

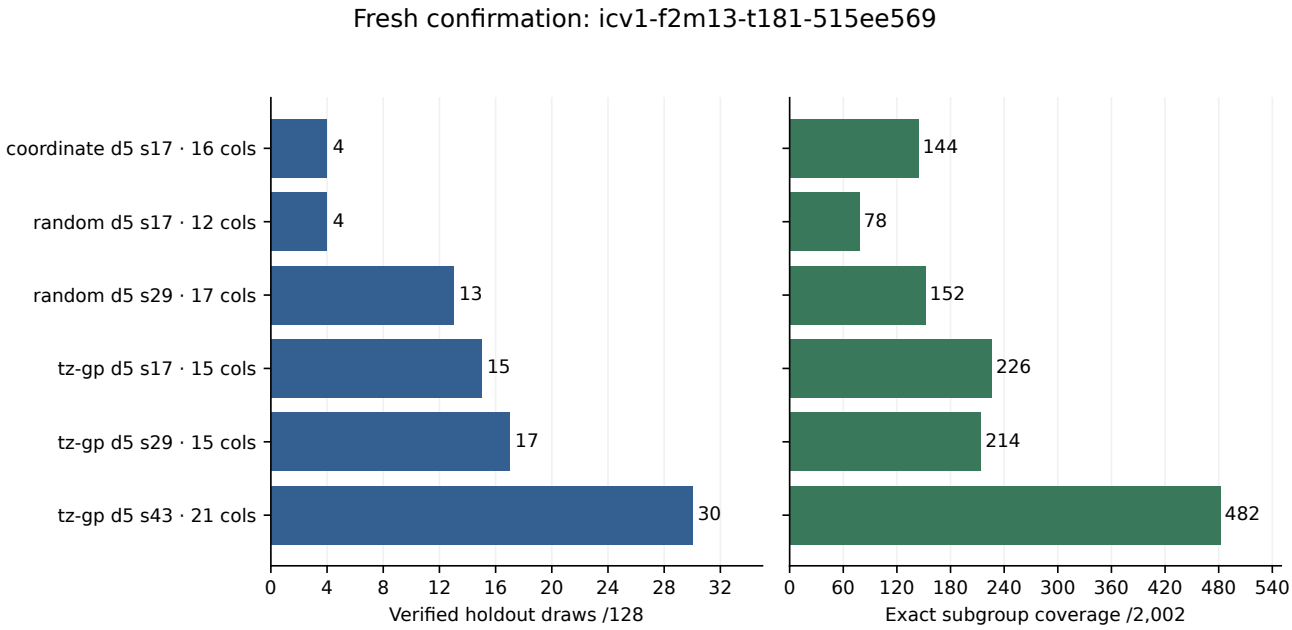
The subgroup gate matters

On **icv1-f2m31-tm90707-c95f16f5**, three coordinate parameterizations and the tested random base support **zero** nonidentity subgroup targets. The tested trace-zero/Frobenius span supports 2 of 1,439,392. Every old whole-curve positive control passed, but every random probe missed. The proper-subfield degree 9 control likewise supports zero of 36 targets. The new CLI requires a finite positive control annihilated by the actual prime subgroup order; earlier evidence keeps its historical scope.



Solid boxes show observed checks; dashed routes remain proposed. Complete native pricing and certified regularity remain unknown.

Fresh confirmation and independent arithmetic



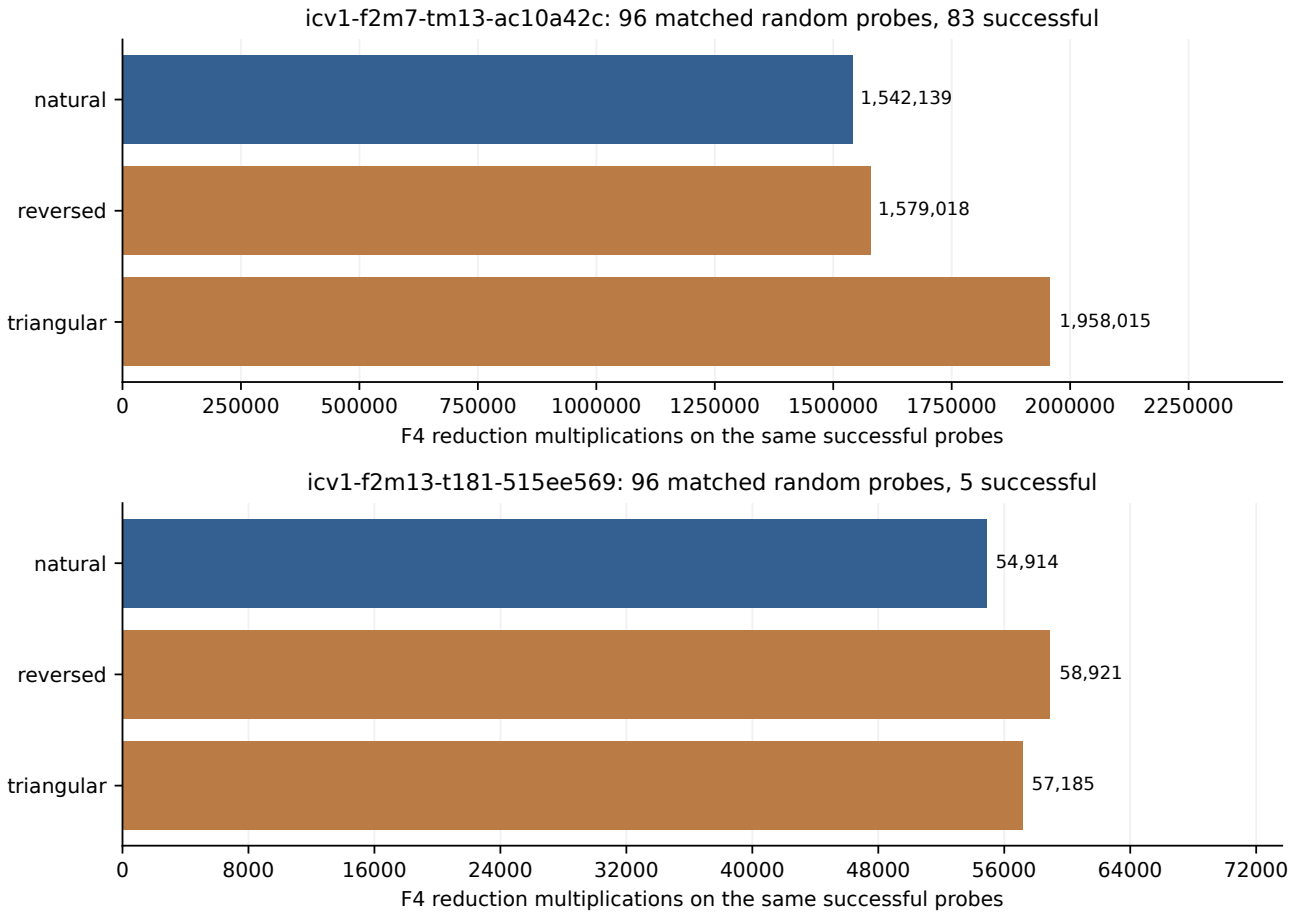
Target seed 2026101802; 122 distinct holdout targets per case. GP seeds17/29 have15 columns; seed43 has21.
Exact support is independently enumerated with a second native arithmetic implementation.
Observed successful solving degree remains3. This is yield/support evidence, not cheap solving or an end-to-end win.

The sampled screen and fresh confirmation are separate, frozen rounds. Constructor seeds 17/29 were retained; seed 43 is a new GP construction. The independent implementation checks the registered polynomial basis, curve equation, recorded generator order, every native truth verdict, and every recorded point-witness sum. It audits 3,923 original trial records and 1006 witnesses on 60 sealed attempts (59 completed, one failed empty base). The failed case remains null, not zero-yield completion.

The exact census enumerates signed point pairs and restricts their sums to the prime subgroup. It is a costly diagnostic, not an available-for-free large-field oracle. This is second-implementation checking on the same host, not an independent-host certificate or an admitted benchmark.

Primary degree 31 is exploratory E0 evidence, with r = 1,439,393 and cofactor 1,492; field degree is not subgroup bit length. The degree 7 fixture uses a=1, while the degree 13 secondary uses a=0. The degree 9 control has a proper intermediate subfield and cannot support a prime-degree conclusion. The degree 31 Frobenius block differs from the irreducible blocks described for degrees 53/83/131 in AGENTS.md. The degree 83 confidence gate and exact challenge remain unperformed. Frobenius-generated prefixes are not automatically Frobenius-stable spaces.

Representation and rank tradeoff



Exact signed point sets and target outcomes match; positive-control work is excluded.
All reversed/triangular aggregates exceed the natural basis. One reversed basis raises training rank10→11.
Neither changed basis passes the preregistered strict per-success work-improvement filter. Total pricing is incomplete.

Changing the basis preserves the exact signed point set and random-probe outcomes, but does not lower aggregate successful F4 work here. All changed bases fail the preregistered strict per-success component-improvement filter. On the degree 7 fixture, reversed order raises training rank 10 to 11 at higher F4 work. The current first-witness policy is therefore a separate lead; more rank alone is not cheaper relation collection.

Follow-up and evidence

The new draft card is XP-20261018-factor-base-shape-witness-leads: cofactor-aware shapes and rank-aware witness selection. It remains unrun, zero-funded and unleased, with no frozen executable protocol. The older card from 1963 is unchanged.

Sources: PROTOCOL.md, FOLLOWUP_PROTOCOL.md, COVERAGE_PROTOCOL.md; observations.json, exact-coverage.json; analysis/src; reports/run-20261018T0437Z and reports/run-20261018T0443Z; validation.md and runs.tar.xz. No accepted wall time, full-pipeline speedup or certified regularity is claimed.